RudraShield

UNIVERSITY CYBER SECURITY VALUE-ADDED PROGRAM (U-CSVAP)

RudraShield Cyber Systems

Industry Integrated Cyber Security Employability Framework

Transforming Students into Industry-Ready Cyber Security Professionals



- Industry Integrated Learning
- Practical Skills and Hands-on Labs
- Industry Projects and Internship Support
- Placement Readiness and Career Support
- Stronger Employability. Better Future.

Building Industry-Ready Talent.
Strengthening University-Industry Collaboration.
Enhancing Graduate Employability.
Creating Future Cyber Security Professionals.

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EXECUTIVE SUMMARY



The University Cyber Security Value-Added Program (U-CSVAP) is an **industry-integrated learning initiative** designed to strengthen student employability, practical skills, and placement outcomes across existing academic programs.

The program is delivered as a **Value-Added / Industry Readiness Program** alongside existing degree programs and does not require modification of the university's approved curriculum.

STRATEGIC OBJECTIVES

STUDENT OUTCOMES

- Develop industry-relevant cyber security skills
- Gain hands-on practical experience
- Work on real-world projects and case studies
- Enhance internship readiness
- Improve employability and placement readiness
- Build professional portfolio and confidence

INSTITUTIONAL OUTCOMES

- Enhance student value proposition and program attractiveness
- Improve placement statistics and graduate employability
- Strengthen industry collaboration and engagement
- Support accreditation, ranking and quality initiatives
- Develop future-ready talent pipelines for industry
- Build institutional reputation and differentiation

PROGRAM HIGHLIGHTS

- Industry Expert Led Training
- Hands-on Laboratories
- Real-world Projects
- Internship Opportunities
- Placement Readiness
- Industry Certificate

Our objective is to empower students with industry-relevant skills, strengthen university outcomes, and create a sustainable ecosystem for cyber security talent development.

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CYBER SECURITY
LEARNING & LAB ECOSYSTEM



Industry-integrated practical learning framework designed to develop technical, operational and professional cyber security competencies across undergraduate and postgraduate programs.

CYBER SECURITY COMPETENCY FRAMEWORK

LEVEL	FOCUS AREA	KEY CAPABILITIES
LEVEL 01 FOUNDATION SKILLS	Build strong fundamentals and understanding of cyber security concepts.	• Cyber Security Fundamentals • Networking Essentials • Linux Fundamentals • Cyber Hygiene & Awareness
LEVEL 02 TECHNICAL SECURITY SKILLS	Develop core technical skills through hands-on learning.	• Web Application Security • Secure Coding Practices • Vulnerability Assessment • Network Security • Operating System Security
LEVEL 03 SECURITY OPERATIONS	Understand security operations and threat management in real-world environments.	• Security Operations (SOC) • Threat Intelligence • Security Monitoring • Incident Response • Log Analysis & SIEM
LEVEL 04 ADVANCED SECURITY	Gain advanced skills to address complex security challenges.	• Cloud Security • Digital Forensics • Malware Analysis • Application Security • Threat Hunting
LEVEL 05 INDUSTRY READINESS	Apply skills in advanced labs, simulations and industry projects.	• Penetration Testing (VAPT) • Red Team Exercises • Blue Team Exercises • Capture The Flag (CTF) • Cyber Range Exercises
LEVEL 06 PROFESSIONAL READINESS	Prepare for careers through real-world exposure and mentorship.	• Industry Projects • Internship Program • Professional Portfolio Development • Interview & Placement Readiness

CYBER SECURITY LAB ECOSYSTEM

World-class labs and virtual environments for hands-on learning and industry readiness.



SOC SIMULATION LAB
Real-time monitoring, log analysis and threat detection



PENETRATION TESTING LAB
Vulnerability assessment and ethical hacking



CLOUD SECURITY LAB
Cloud configurations, security controls and compliance



DIGITAL FORENSICS LAB
Forensic analysis, data recovery and investigation



SECURE CODING LAB
Hands-on secure coding and code review



THREAT INTELLIGENCE LAB
Threat research, analysis and intelligence feeds



CYBER RANGE
Simulated enterprise environment for attack and defense scenarios



CTF PLATFORM
Gamified challenges to enhance problem solving and practical skills

STUDENT LEARNING JOURNEY

LEARN → Conceptual learning with industry-aligned content

PRACTICE → Hands-on labs to build practical technical skills

SIMULATE → Simulated environments and cyber range exercises

DEFEND → SOC operations and defensive security practices

ATTACK → Ethical hacking and realistic attack simulations

DEPLOY → Industry projects, internships and career readiness

→ **Industry-Ready Cyber Security Professional**

STUDENT OUTCOMES

- **300+** Hours of Hands-on Labs
- **50+** Real-world Security Scenarios
- **10+** Industry Projects & Case Studies
- **CTF** Challenges & Competitions
- **Internship** Support & Industry Mentorship
- **Placement** Readiness & Career Support
- **Professional** Portfolio & Industry Certification

Our program ensures students gain practical skills, industry exposure and professional readiness to excel in the dynamic field of cyber security.

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DELIVERY & GOVERNANCE MODEL

A well-structured delivery and governance framework ensures academic alignment, quality execution, and measurable outcomes.



PROGRAM DELIVERY MODEL

Industry Expert-Led Sessions
Interactive sessions by cyber security practitioners and domain experts.

Hands-on Lab Experience
Extensive practical exposure in specialized labs and virtual environments.

Real-world Projects & Case Studies
Industry-aligned projects that build problem-solving and application skills.

Capture The Flag (CTF) Challenges
Gamified cyber range exercises to enhance attack-defense skills.

Internship & Industry Exposure
Guided internship support and industry interaction opportunities.

Career & Placement Readiness
Resume building, interview preparation and placement support.

> "Blending academic rigor with industry practice to build future-ready cyber security professionals."

GOVERNANCE FRAMEWORK

A joint governance committee ensures effective implementation, continuous improvement and outcome tracking.

UNIVERSITY REPRESENTATIVES	JOINT PROGRAM GOVERNANCE COMMITTEE	INDUSTRY REPRESENTATIVES
Dean Academics	Collaborative oversight for academic alignment, quality assurance, student outcomes and industry engagement.	Program Director
Head / Director (Departments)		Technical Lead / Architects
Faculty Coordinators		Industry Mentors
Training & Placement Cell		Placement & Outreach Lead
IQAC Representative		Subject Matter Experts

COMMITTEE RESPONSIBILITIES

- Review program progress and outcomes
- Ensure academic alignment and quality
- Approve industry projects and initiatives
- Monitor student performance
- Plan placements and industry engagement

QUALITY ASSURANCE & CONTINUOUS IMPROVEMENT

- Regular assessments and feedback
- Data-driven program performance analysis
- Stakeholder feedback (Students, Faculty, Industry)
- Continuous curriculum enhancement
- Outcome tracking and impact measurement

Our delivery and governance model ensures a seamless blend of academic excellence and industry relevance, creating measurable value for students and the institution.

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PARTNERSHIP & ENGAGEMENT MODEL

A flexible and sustainable partnership model designed to create **long-term value** for the university, students, and industry.



PARTNERSHIP FRAMEWORK

Collaborative Partnership
A strategic collaboration aligned with the university's academic goals and student outcomes.

No Curriculum Change
Delivered as a Value-Added / Industry Readiness Program alongside existing degree programs.

Co-created for Impact
Designed jointly with faculty and industry to ensure relevance, rigor and real-world applicability.

Outcome-Driven Collaboration
Focused on measurable outcomes in skills, placements, certifications, and industry engagement.

Long-term Sustainability
Continuous innovation, updates and support to keep pace with evolving industry needs.

COMMERCIAL ENGAGEMENT MODEL

FLEXIBLE ENGAGEMENT OPTIONS

ANNUAL PROGRAM MODEL
Annual subscription per student for access to training, labs, projects, certifications and placement support.

COHORT-BASED MODEL
Structured engagement for specific student cohorts or batches.

VOLUME-BASED PRICING
Scalable pricing based on number of students and selected program components.

CUSTOMIZED PACKAGES
Modular options to choose the right mix of labs, certifications, projects and services.

WHAT'S INCLUDED

- Training & Mentorship
- Lab Access & Cyber Range
- Certifications & Assessments
- Projects & Case Studies
- Placement & Career Support

WHY PARTNER WITH US?

ENHANCE STUDENT VALUE
Equip students with industry-relevant skills, practical experience and professional confidence.

IMPROVE PLACEMENT OUTCOMES
Better employability, higher placement rates and stronger recruiter confidence.

STRENGTHEN INDUSTRY CONNECT
Deepen engagement with industry through projects, mentors, guest sessions and more.

SUPPORT ACCREDITATION & RANKINGS
Contribute to key metrics for accreditation, NIRF, NBA and other rankings.

BUILD INSTITUTIONAL REPUTATION
Position the university as a leader in cybersecurity education and innovation.

FUTURE-READY ECOSYSTEM
Create a sustainable ecosystem for continuous learning, research and talent development.

Together, we can build a future-ready cybersecurity talent pipeline, drive impactful outcomes, and create lasting value for students and the institution.

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INSTITUTIONAL VALUE & STUDENT READINESS

Enriching academic quality and building future-ready cyber security professionals through practical learning, skill development and professional preparation.



IMPACT AREAS

ACADEMIC EXCELLENCE

- Outcome-oriented cyber security education
- Practical and application-based learning
- Structured assessments for skill mastery
- Continuous academic enrichment

STUDENT DEVELOPMENT

- Hands-on laboratory experience
- Real-world simulations and exercises
- Technical competency development
- Strong foundation for advanced learning

PROFESSIONAL READINESS

- Application-oriented knowledge and skills
- Problem-solving and analytical abilities
- Assessment and certification readiness
- Professional communication skills

APPLIED LEARNING FRAMEWORK

- Cyber range and lab environments
- Case-study driven learning
- Project-based assessments
- Capture The Flag (CTF) challenges

INSTITUTIONAL DIFFERENTIATION

- Future-ready academic offerings
- Enhanced student value proposition
- Strengthened academic portfolio
- Modern, industry-aligned learning ecosystem

STUDENT READINESS OUTCOMES

300+
Hours of Practical Laboratory Exposure

50+
Real-world Security Simulations

CTF
Challenges to Build Offensive & Defensive Skills

Project-Based Learning
To Solve Real-world Security Problems

Certification Readiness
Preparation for Industry-recognized Certifications

Career Preparedness
Confident, Skilled and Future Ready

INSTITUTIONAL BENEFITS

Enhanced Academic Quality
Strengthen curriculum with practical, skill-based cyber security education.
01

Elevated Student Experience
Engaging, hands-on learning that builds confidence and competence.
02

Future-Ready Graduates
Well-prepared for the evolving demands of the digital world.
03

Stronger Institutional Reputation
Position the university as a leader in emerging technology education.
04

Sustainable Academic Growth
Long-term value through relevant programs and future-focused initiatives.
05

Together, we empower institutions to build a strong cyber security learning ecosystem that develops **skilled, confident and future-ready professionals.**